

Problem 1.3 - Minimum-Time Broadcast

1. Method

The 32-vertex, degree-3 graph was represented in NetworkX using the 32-node outer cycle and the 16 cross-circle links. Starting from source vertex 1, a breadth-first search (BFS) was used to explore reachable informed-vertex states. At each state, the routine generated the maximum-size legal set of simultaneous transmissions: a sender had to be informed at the start of the step, could send to at most one uninformed neighbor, and each receiver could be used at most once. BFS then explored the resulting states in increasing step depth until all 32 vertices were informed.

2. Total Number of Steps

S = 6 steps. The program found a valid broadcast schedule that informs all 32 vertices in six communication steps.

3-4. Active Links and Number of Edges per Step

Step	Active links	Edges
1	1-2	1
2	1-32, 2-3	2
3	1-12, 2-23, 3-4, 32-31	4
4	3-19, 4-5, 12-11, 23-22, 31-30, 32-16	6
5	4-26, 5-6, 11-10, 12-13, 16-15, 19-18, 22-21, 23-24, 30-29, 31-9	10
6	15-25, 16-17, 18-7, 19-20, 21-14, 24-8, 26-27, 29-28	8

Total edge uses: $1 + 2 + 4 + 6 + 10 + 8 = 31$. This is consistent with the fact that one source begins informed and each of the other 31 vertices must receive the message once.

Why the Theoretical Lower Bound Is 5

In one communication step, each informed processor can inform at most one new processor. Therefore the number of informed processors can at most double each step. After S steps, at most 2^S processors can be informed. For 32 processors, $2^S \geq 32$, so $S \geq 5$.

A five-step solution would therefore require perfect doubling at every step: **1 -> 2 -> 4 -> 8 -> 16 -> 32**. The verification routine explored all perfect-doubling states permitted by the graph and found that this sequence cannot be maintained because the graph topology forces overlap in available receivers. Thus five steps is impossible. Since a valid six-step schedule exists, the minimum broadcast time is **S = 6**.

Brief Pseudocode

```
CREATE the 32-vertex graph
VALIDATE 32 vertices, 48 edges, degree 3, and connectivity

source <- vertex 1
queue <- {(source, empty schedule)}
visited <- {source}

WHILE queue is not empty:
    REMOVE the oldest state from queue
    IF all 32 vertices are informed:
```

```
    RETURN its schedule

FIND all maximum-size legal simultaneous transmissions
FOR each transmission choice:
    FORM the next informed-vertex state
    IF state has not been visited:
        ADD the transmission step to its schedule
        ADD state to visited
        APPEND state to queue

VERIFY the 5-step lower bound separately:
    REQUIRE perfect doubling 1 -> 2 -> 4 -> 8 -> 16 -> 32
    EXPLORE all graph-valid perfect-doubling states
    IF none reaches 32 vertices:
        5 steps is impossible

CONCLUDE minimum broadcast time  $S = 6$ 
```